

Indian Institute of Technology Kharagpur

CS60111 GIS · TERM PROJECT · GROUP 8

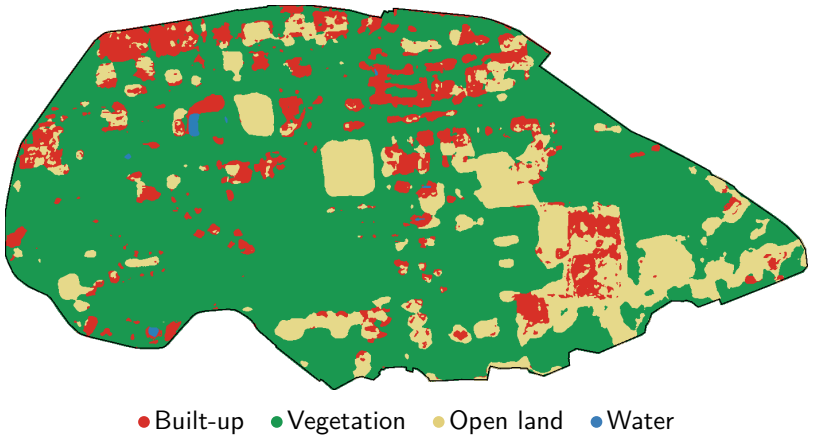
Land Use / Land Cover Mapping and Change Detection

IIT Kharagpur campus, 2020 vs 2025, from multi-temporal Sentinel-2

Ashutosh Sharma 23CS10005
Krishnkant Sahu 23CS10035
Sanskar Sovitkar 24CS10131

Term project evaluation, Nov 2026

2025 land cover map, Random Forest, 10 m



Can we measure how our own campus changed?

RESEARCH QUESTION

How much of the 469 ha campus changed land cover between the 2019-20 and 2024-25 dry seasons, and from what to what?

Why a campus? Global products (Dynamic World, ESA WorldCover) map at 10 m but are tuned for regions. A 469 ha site tests whether they hold at the scale where planners actually decide: one construction compound, one lawn, one pond.

01 **Map 4 classes at 2 dates**
Sentinel-2, one Random Forest

02 **From-to change table in ha**
30 m minimum change width

03 **Accuracy against independent points**
WorldView-2 and -3, area-weighted

04 **Reproducible pipeline**
one Earth Engine script, one Python runner

New in this version: yearly dating of the south-east compound · GLCM texture test · is the change real? error-adjusted significance test · change flow diagram · gross gain, loss and swap per class · map area vs point-based area with 95% CI · model tuning log with a held-out half · feature importance · WorldView validation design

A 469 ha fenced campus, mapped at 10 m



Campus boundary: OpenStreetMap way 52435606 (184 vertices). Imagery: Google Earth, Airbus.

469.2 ha
MAIN FENCED CAMPUS

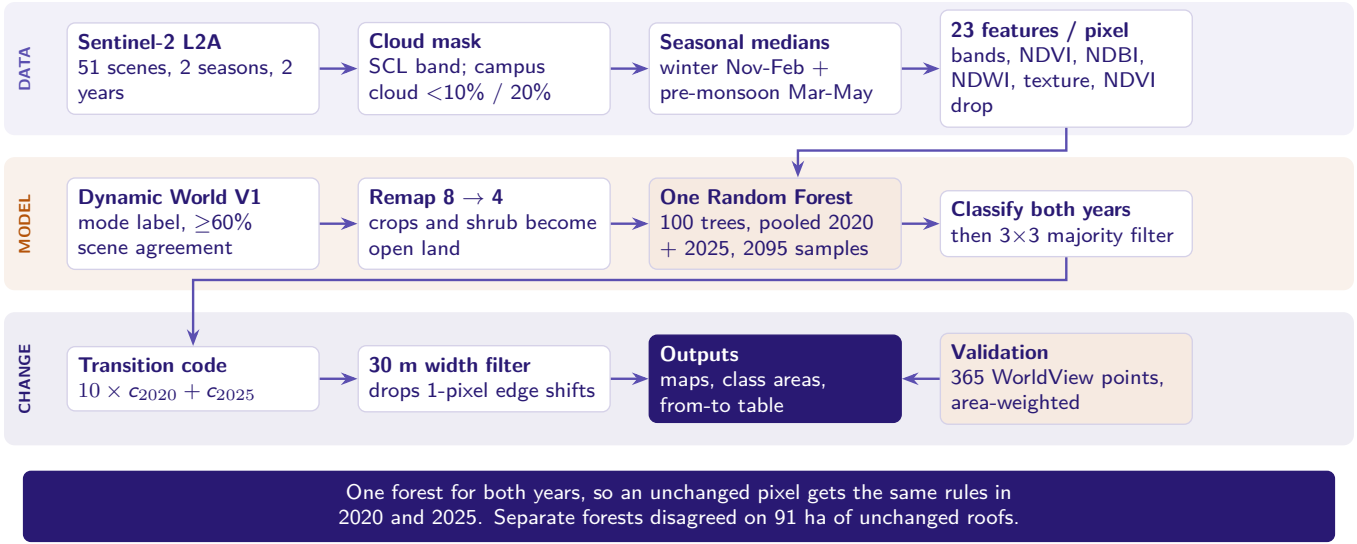
10 m
SENTINEL-2 PIXEL SIZE

45QWE
ONE TILE COVERS ALL OF IT

UTM 45N
EPSG:32645

Halls and quarters outside the fence are not in the study area.

From 51 satellite scenes to a from-to change table



One forest for both years, so an unchanged pixel gets the same rules in 2020 and 2025. Separate forests disagreed on 91 ha of unchanged roofs.

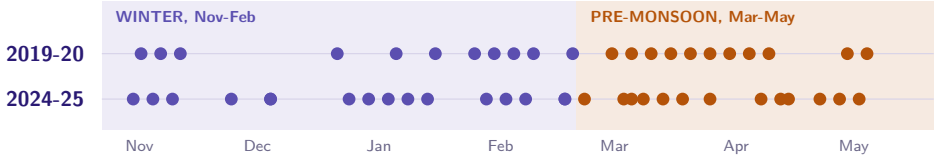
Same seasons in both years, so seasonal greening is not counted as change



Winter 2019-20 median composite, 11 scenes

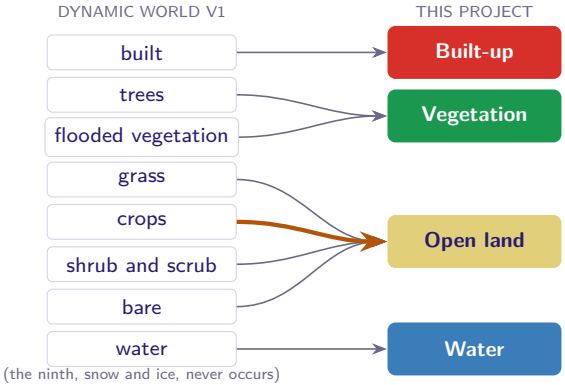


Winter 2024-25 median composite, 16 scenes

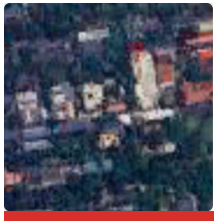


Every dot is one scene that passed the campus cloud test. By April lawns dry out but tree canopy stays green, so the pre-monsoon season helps split them.

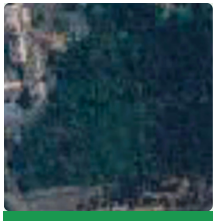
Eight Dynamic World labels collapse into four campus classes



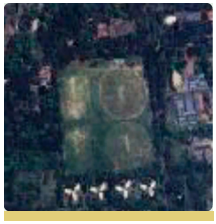
The crops rule. Dynamic World calls about 50 ha of campus lawns *crops*. There is no farmland inside the fence, so crops become open land. Calling them vegetation would hide open-to-built conversions.



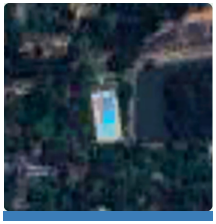
Built-up
roofs, roads, courts



Vegetation
canopy, dense woody



Open land
lawns, fields, bare plots



Water
ponds, tanks

23 features per pixel; the seasonal and texture ones earn their place

10	winter bands B2-B8A, B11, B12
3	winter NDVI, NDBI, NDWI
2	texture: 3×3 std of NDVI, B8
7	pre-monsoon: 4 bands, 3 indices
1	NDVI drop, winter minus pre-monsoon

= 23 features

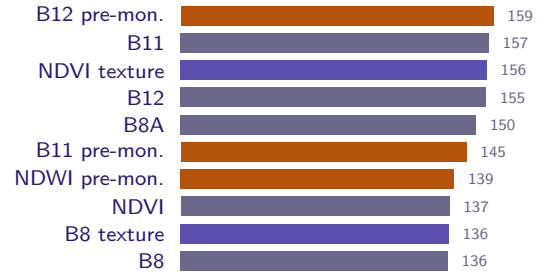
$$\mathrm{NDVI} = \frac{B8 - B4}{B8 + B4}$$

$$\mathrm{NDBI} = \frac{B11 - B8}{B11 + B8}$$

$$\mathrm{NDWI} = \frac{B3 - B8}{B3 + B8}$$

Greenness, built-up and water, each separating a pair of classes that look alike in raw bands.

RANDOM FOREST IMPORTANCE, TOP 10 OF 23 (GINI, EARTH ENGINE)



The **top feature is pre-monsoon SWIR (B12)**, and **5 of the top 10** are seasonal or texture features added in the last fix. Short-wave infrared is bright on roofs and dry soil, dark on moist leaves.

Tuning the model: three runs, each fix aimed at one measured error

RUN 1

Baseline

Equal samples per class, one forest per year.

Errors: built-up 224 ha (Dynamic World says ≈110); 91 ha built → vegetation on unchanged roofs; zero open-land samples in 2020.

RUN 2

Fix the sampling

Samples near real class shares (300 / 700 / 150 / 40); **one forest pooled over both years**; crops → open land; 30 m change filter.

Left over: open land only 32 ha; points say about 146.

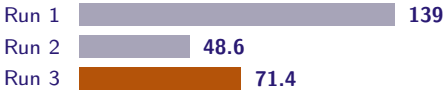
RUN 3 (FINAL)

Fix open land

Add pre-monsoon composite, 3×3 texture, shrub → open land, 450 open samples.

Result: open land 2020 at 103 ha; OA 0.61 → 0.67 (2020), 0.74 → 0.77 (2025).

CHANGED AREA REPORTED (HA)



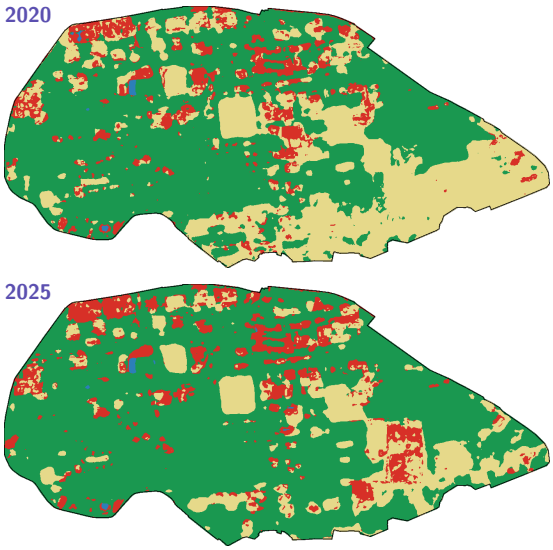
Held-out check, so tuning does not overfit the points. Each change was chosen on the odd-numbered validation points and scored on the even-numbered half it never saw. Mean OA of both years, first version → final:

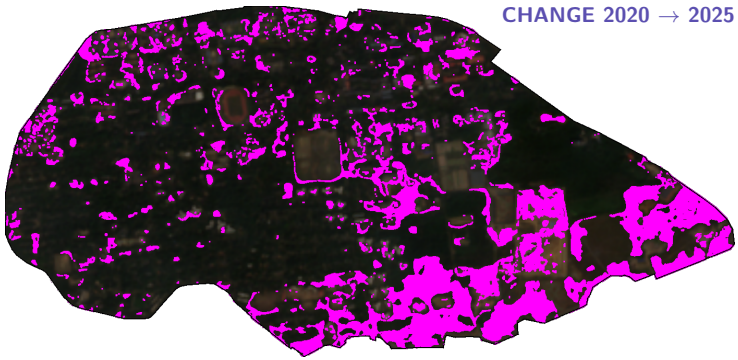
odd half (chosen on) 0.65 → **0.685** **even half (held out)** 0.70 → **0.76**

Run 2 to 3 raises change from 48.6 to 71.4 ha because more open land is now mapped, so more of it can be seen turning green.

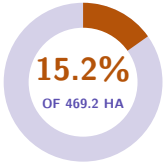
GLCM texture, tried after (NIR 3×3 or 5×5, NIR + NDVI): 0.645-0.67 on the odd half, below 0.685, so not adopted.

Vegetation grew, open land shrank: 15.2% of the campus changed





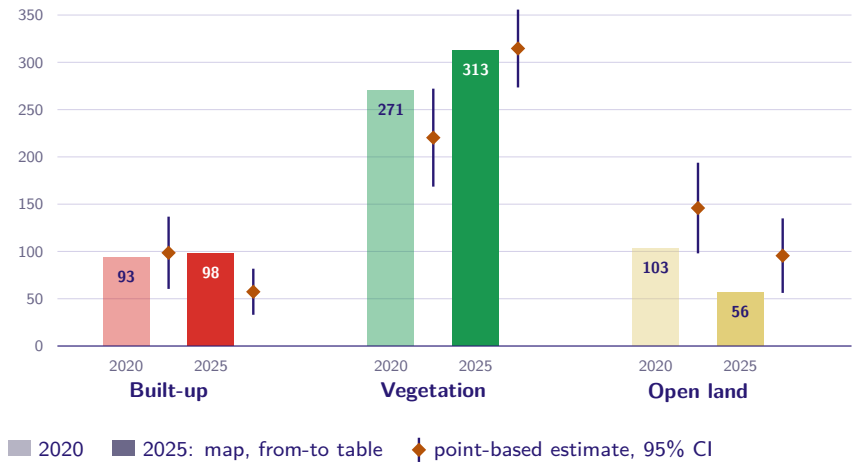
Magenta: changed patches at least 30 m wide, over the 2025 image



71.4 ha CHANGED, ≈14 HA / YEAR

Open land in the east and south-east turns green or red. Same forest, same 23 features in both years, so the difference is the ground, not the model. Before the 30 m width filter the change map holds 103.3 ha; the 31.9 ha it removes is mostly a one-pixel ring around roofs.

Vegetation +42.2 ha, open land −46.8 ha; the points agree on both



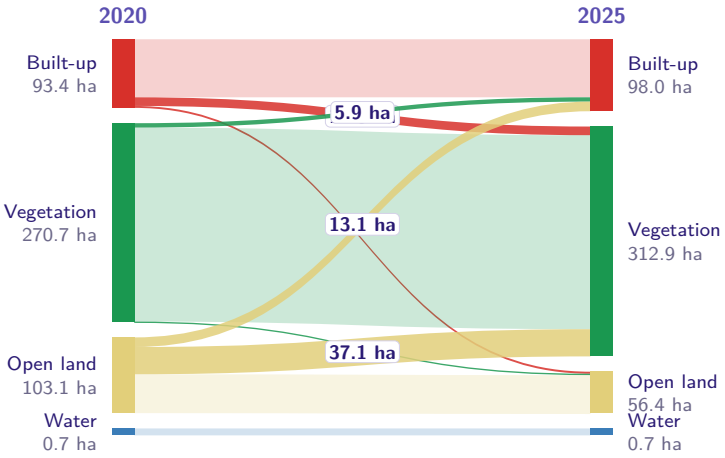
Vegetation +42.2
gained 48.9, lost 6.7 ha

Open land −46.8
gained 3.4, lost 50.2 ha

Built-up +4.6
gained 19.0, lost 14.5 ha

Map vs points. The point estimate (Olofsson et al. 2014) corrects for known map errors. Mapped open land sits 43 ha (2020) and 39 ha (2025) below it. Built-up: map +4.6, points -41 ± 45 ha; neither is distinguishable from zero. Only vegetation's intervals do not overlap (169-272 vs 273-356 ha).

Open land feeds both vegetation and building; built-up mostly swaps place



Band width = hectares; faint bands stayed the same class. From-to table after the 30 m filter.

ha	gained	lost	net	swap
Built-up	19.0	14.5	**+4.6**	29.0
Vegetation	48.9	6.7	**+42.2**	13.3
Open land	3.4	50.2	**−46.8**	6.9

Swap = 2 × min(gain, loss): area that changed place without changing the class total (Pontius et al. 2004).

37.1 **Open → vegetation**
east-end grassland growing into scrub

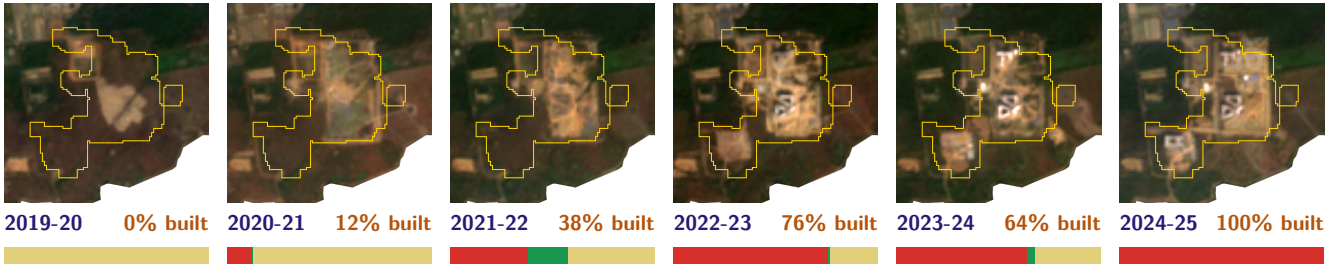
13.1 **Open → built-up**
south-east construction compound

11.8 **Built-up → vegetation**
mostly canopy closing over roofs

5.9 **Vegetation → built-up**
new blocks in the south

Built-up looks stable, but is not. Net +4.6 ha, yet 33.5 ha of it changed, 29.0 of that swap. Part of the loss is canopy closing over roofs, so some swap is the 10 m pixel, not new ground. Vegetation and open land change one way.

The south-east compound was cleared in 2020 and had roofs by early 2023



Winter true colour, site in yellow. Bar: the forest's class for the site's open → built-up pixels ● built-up ● vegetation ● open land

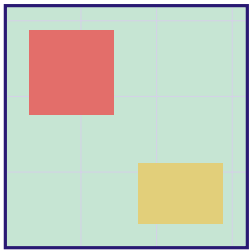
Method
The forest trained on 2020 and 2025 classifies every winter in between, with its pre-monsoon pair. Site: the largest open → built-up cluster, 10.8 of the campus's 13.1 ha, inside a 24 ha outline.

Cleared in 2020, roofed by 2023
A bare patch shows by early 2020 and the whole site is bare in 2020-21. Roof shapes appear in 2022-23; by 2024-25 a second block stands to the south-west.

The map runs ahead of the roofs
38% "built" in 2021-22 is bare soil, the roof-like SWIR signal from the accuracy slide. The 2024 dip is year-to-year noise; 0% and 100% at the ends hold by definition.

Validation labels come from WorldView, 20 times sharper than Sentinel-2

≈400 WorldView pixels

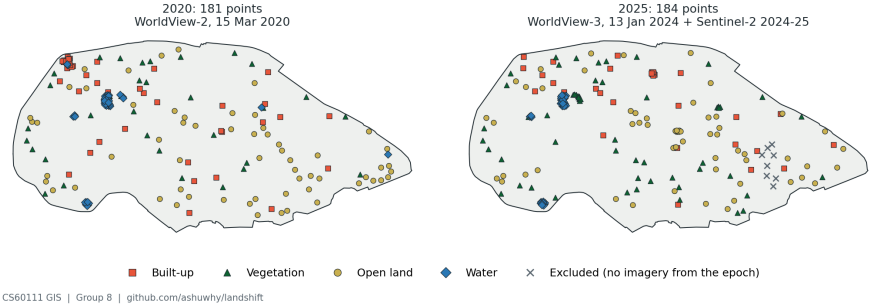


one Sentinel-2 pixel = 10 m
grid: 0.5 m, WorldView pan scale

A 10 m pixel that mixes a roof and a lawn is one label to the classifier but readable by eye on WorldView.

Stratify by map class, 25+ points per class per year → **Label by eye** WorldView-2 (15 Mar 2020), WorldView-3 (13 Jan 2024) → **Offset-correct** to the Sentinel-2 grid → **Area-weight** OA, user's, producer's, area ± 95% CI

Independent validation points, labelled on high-resolution imagery



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365 points: 181 for 2020, 184 for 2025, labelled by Krishnkant Sahu. Independent of Dynamic World, unlike the 883 held-out training pixels.

Accuracy 0.67 (2020) and 0.77 (2025); open land is the weak class

OVERALL ACCURACY 2020

0.67

95% CI 0.57 to 0.78

OVERALL ACCURACY 2025

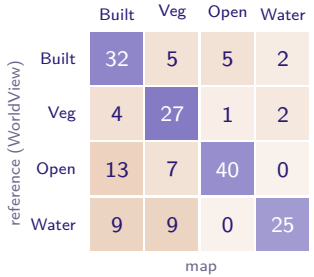
0.77

95% CI 0.67 to 0.87

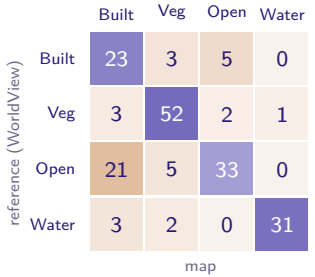
Area-weighted (Olofsson et al. 2014). Against Dynamic World's own held-out pixels the map scores 0.82 / 0.88 (kappa 0.73 / 0.81), but that only measures agreement with the training source.

Open land: producer's accuracy 0.41 and 0.33, so most real open land is mapped as built-up or vegetation. Vegetation is found 92% of the time in both years.

2020: OA 0.67 (0.57 to 0.78)



2025: OA 0.77 (0.67 to 0.87)

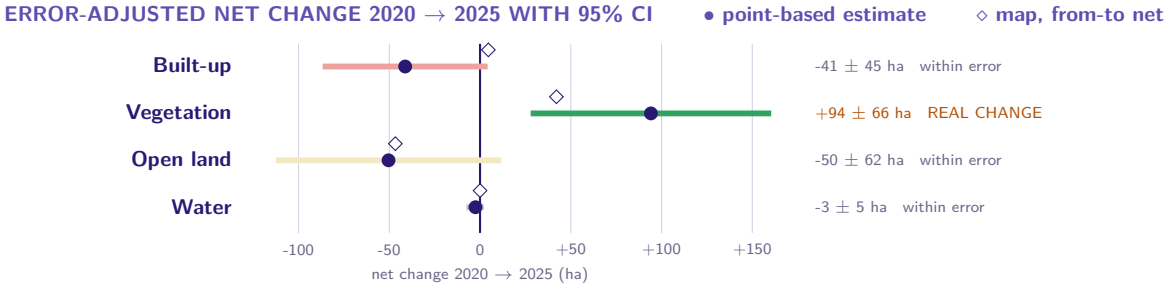


Point counts. Rows: WorldView reference; columns: map. Shade: share of the row; indigo diagonal is correct, saffron is error.

Open → built-up
13 of 60 points in 2020, 21 of 59 in 2025. Bare plots and dry courts share the SWIR signal of roofs.

Water → built-up or vegetation
18 of 43 water points in 2020 sit on pond edges, where a 10 m pixel is half bank.

Only the vegetation gain is bigger than our map error



Method
Olofsson adjusted area per class, per year, from the WorldView points. The 2020 and 2025 point sets are independent, so the half-width of the difference is $\sqrt{h_{2020}^2 + h_{2025}^2}$.

Vegetation +94 ± 66 ha
The whole interval sits above zero ($z = 2.8$). Campus greening is the one change we can defend against map error.

Built-up and open land
Intervals cross zero (-41 ± 45, -50 ± 62 ha). Halving them needs about $4\times$ the points (CI $\propto 1/\sqrt{n}$), so that is where new labelling should go.

15.2% of the campus changed, mostly open land turning green

71.4 ha
changed: 37.1 open → vegetation, 13.1 open → built-up (south-east compound), 11.8 built-up → vegetation

+42.2 / −46.8 ha
net, after the 30 m filter: vegetation gained 48.9 and lost 6.7 ha; open land lost 50.2 and gained 3.4 ha

33.5 ha
of built-up changed hands while its net moved under 5 ha

0.67 / 0.77
overall accuracy on 365 independent WorldView points

Yearly composites date the biggest site: cleared in 2020, roofed by 2023.

WHAT THE MAP CANNOT SUPPORT

- **Open land is under-counted.** Producer's 0.41 and 0.33; the map sits about 40 ha below the points. **So:** treat its area as a lower bound.
- **Change accuracy is not measured.** The change map combines two maps at 0.67 and 0.77. **So:** read 71.4 ha as an estimate.
- **10 m pixels mix surfaces.** Roofs under canopy blur; GLCM texture did not help; six bands are 20 m. **So:** small features are unreliable.
- **The 30 m filter drops narrow change.** It cuts 103.3 ha of raw change to 71.4. **So:** a single new building on a lawn can be missed.
- **Labels come from Dynamic World.** It calls lawns crops (50 ha in 2020); we remap by rule. **So:** the independent points check that rule.
- **Few points, tiny water.** Overall-accuracy intervals are about ±0.10; water is 0.7 ha. **So:** water area is not estimable.

WHAT WE WOULD DO NEXT

Own labels	Change accuracy	4× the points	3 m imagery
train on WorldView labels, not Dynamic World	sample the change map itself; change ± CI	halve the built-up and open-land intervals	roofs under canopy, where 10 m texture failed

Who did what

Ashutosh Sharma
23CS10005

- Earth Engine pipeline, first drafts to `gee/lulc_pipeline.js`
- Diagnosed run 1 and ran the tuning: proportional sampling, one forest, crops rule, 30 m filter, pre-monsoon and texture
- Area-weighted accuracy and area estimates (Olofsson, Stehman)
- GLCM texture test; yearly composites dating the south-east compound
- Python runner that reproduces every number; all figures, README and this deck

Krishnkant Sahu
23CS10035

- Independent validation set: 365 points, stratified on map classes
- Labelled on WorldView-2 and -3, offset-corrected to Sentinel-2
- Labelling notes: dates, rules, hard cases
- Point layers, sampling and export scripts

Sanskar Sovitkar
24CS10131

- Study-area figure: Google Earth Pro view with OSM boundary, north arrow, scale bar
- Literature review of Sentinel-2 / Random Forest change studies
- Framed the research gap: global products vs one campus

Code and results: **github.com/ashuwhy/landshift** · gee/lulc_pipeline.js · scripts/run_pipeline.py · slides/make_charts.py

Thank you

15.2% of the campus changed,
mostly open land turning green.

github.com/ashuwhy/landshift

Ashutosh Sharma
Krishnkant Sahu
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